

3B Window Functions

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Planned Approach

First, I will find a dataset that includes daily time series data for at least two items, such as stock prices or cryptocurrency prices starting from January 1, 2022. The dataset will include the item name, date, and closing price.

Next, I will clean and organize the data. This includes making sure the dates are formatted correctly and that the data is sorted in order from earliest to latest. Window functions only work properly when the data is ordered correctly.

Then, I will calculate the year-to-date (YTD) average for each item. This means I will group the data by item and year and compute the running average from the start of each year to each date.

After that, I will calculate the six-day moving average for each item. This will take the current day and the previous five days to compute a rolling average. This helps smooth out short-term changes in the data.

Finally, I will review the results and compare the original values to the calculated averages.

Anticipated Data Challenges

One challenge may be missing dates, such as weekends or holidays in stock data. While this does not stop the calculations, it can affect how the moving average looks.

Another challenge is making sure the data is correctly grouped and ordered. If the data is not sorted by date, the calculations may be incorrect.

If the dataset covers multiple years, I will need to make sure the year-to-date average resets at the beginning of each year.